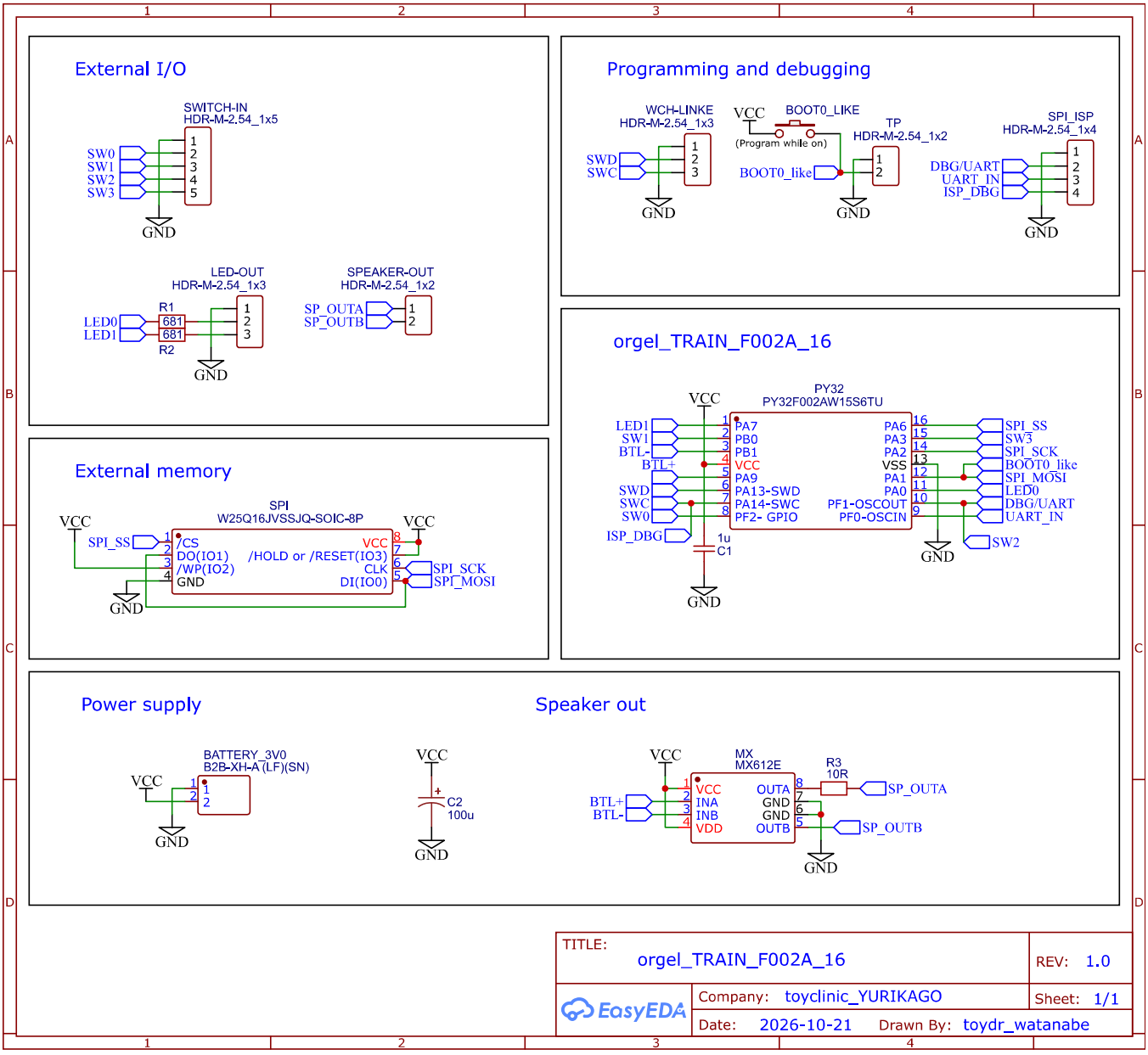

External I/O
SWITCH-IN
HDR-M-2.54_1x5
SW0
SW1
SW2
SW3
GND
LED-OUT
HDR-M-2.54_1x3
R1
681
LED0
LED1
681
R2
GND
SPEAKER-OUT
HDR-M-2.54_1x2
SP_OUTA
SP_OUTB
Programming and debugging
WCH-LINKE
HDR-M-2.54_1x3
SWD
SWC
GND
VCC
BOOT0_LIKE
(Program while on)
BOOT0_like
TP
HDR-M-2.54_1x2
GND
SPI_ISP
HDR-M-2.54_1x4
DBG/UART
UART_IN
ISP_DBG
GND
External memory
SPI
W25Q16JVSSJQ-SOIC-8P
VCC
SPI_SS
/CS
DO(IO1)
/WP(IO2)
GND
/HOLD or /RESET(IO3)
CLK
DI(IO0)
SPI_SCK
SPI_MOSI
GND
orgel_TRAIN_F002A_16
PY32
PY32F002AW15S6TU
VCC
LED1
SW1
BTL-
BTL+
SWD
SWC
SW0
ISP_DBG
PA7
PB0
PB1
VCC
PA9
PA13-SWD
PA14-SWC
PF2- GPIO
1u
C1
GND
PA6
PA3
PA2
VSS
PA1
PA0
PF1-OSCOUT
PF0-OSCIN
SPI_SS
SW3
SPI_SCK
BOOT0_like
SPI_MOSI
LED0
DBG/UART
UART_IN
SW2
GND
Power supply
BATTERY_3V0
B2B-XH-A(LF)(SN)
VCC
GND
C2
100u
Speaker out
MX
MX612E
VCC
BTL+
BTL-
VCC
INA
INB
VDD
OUTA
GND
GND
OUTB
R3
10R
SP_OUTA
SP_OUTB
GND
TITLE:
orgel_TRAIN_F002A_16
REV: 1.0
EasyEDA
Company: toyclinic_YURIKAGO
Sheet: 1/1
Date: 2026-10-21
Drawn By: toydr_watanabe